

Limitations and Improvements on CLIP

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Project Overview

This project investigates the strengths and weaknesses of visual language model CLIP, and compares them with newer models

- [VideoChat: chat-centric video understanding](#)
- [SigLIP 2: Multilingual Vision-Language Encoders with Improved Semantic Understanding, Localization, and Dense Features](#)
- [PaliGemma 2: A Family of Versatile VLMs for Transfer](#)

The main goal is to understand where CLIP performs well and where it fails, particularly in tasks requiring object relationships, prompt robustness, localization, out-of-distribution recognition, temporal reasoning, visual question answering, and counting.

CLIP is powerful because it learns a shared embedding space between images and text. This allows it to perform zero-shot classification by comparing an image against multiple text prompts. However, the experiments show that this global image-text matching approach has important limitations. CLIP often recognizes objects correctly but struggles when the task requires deeper semantic understanding, such as identifying who is acting on whom, understanding temporal direction, localizing specific objects, or interpreting handwritten digits.

Overall, the project demonstrates that CLIP is useful for broad image-text alignment, but it is not a complete visual reasoning system. Newer models improve some of these weaknesses, but they also introduce new limitations.

We divided the project into three main sections:

Section 1: CLIP Limitations

In this section we discussed where CLIP fails using a few controlled experiments. Since CLIP is a zero-shot model, we scraped images from the internet and used it for the experiment.

Experiment 1: Object Relationship Failure in CLIP

The first experiment tested whether CLIP can distinguish between similar prompts involving a dog and a cat. The prompts included phrases such as “a dog chasing a cat,” “a cat chasing a dog,” “a dog next to a cat,” and “a dog and a cat in the grass.”

The results show that CLIP can recognize that the image contains a dog and a cat, but it has difficulty understanding the exact relationship between them. For example, in one case CLIP assigned high probabilities to both “a dog chasing a cat” and “a cat chasing a dog.” These two prompts contain almost the same nouns, but the semantic meaning is opposite because the subject and object are reversed.

This suggests that CLIP behaves partly like a bag-of-words image-text matcher. It is sensitive to the presence of important words such as “dog,” “cat,” and “grass,” but it is less reliable when the correct answer depends on word order, action direction, or relational structure. Therefore, CLIP can identify visual objects but may fail at compositional reasoning.

Experiment 2: Prompt Sensitivity in CLIP

The second experiment tested whether changing the wording of the prompt changes CLIP’s prediction. The same image and answer choices were evaluated using several prompt templates, such as:

- “Question: {} Answer: {}”
- “Q: {} A: {}”
- “The answer to '{}' is '{}”
- “This image shows '{}”

For one image, CLIP was stable across all prompt templates and produced a prompt consistency score of 1.0. This means that for that example, CLIP’s prediction did not change when the wording changed.

However, for another image, the consistency score dropped to 0.75. In that case, CLIP changed between “a dog chasing a cat” and “a cat chasing a dog” depending on the prompt template. This is important because the image itself did not change; only the wording changed. Therefore, the model’s answer was not fully determined by visual understanding alone.

This result shows that CLIP is prompt-sensitive. Its predictions depend not only on the image but also on the exact natural-language phrasing used in the text prompts. This is a limitation for real-world applications because users may phrase the same question in different ways, and a reliable model should ideally produce consistent answers.

Experiment 3: Localization Limitation Using Grad-CAM

The Grad-CAM experiment examined whether CLIP focuses on the correct visual regions when prompted with labels such as “a dog,” “a cat,” or “a dog and a cat.” The goal was to understand whether CLIP’s prediction is supported by correct spatial attention.

This experiment is important because high classification confidence does not always mean correct visual grounding. A model may output the right label while looking at irrelevant

background regions or only part of the object. Since CLIP was trained mainly for global image-text similarity, it is not explicitly optimized for pixel-level localization.

The Grad-CAM results support the idea that CLIP's visual grounding can be weak. CLIP may recognize the general scene but does not always provide precise localization of the object or relationship described in the prompt. This is one reason why CLIP struggles with tasks that require spatial precision, object counting, segmentation, or relationship reasoning.

Experiment 4: CLIP on Out-of-Distribution MNIST

The MNIST experiment tested CLIP on handwritten digit recognition using zero-shot text prompts. This is an out-of-distribution task because CLIP was trained mostly on natural image-text pairs from the web, while MNIST contains simple grayscale handwritten digits.

The results show poor zero-shot performance. CLIP achieved about 22.7% accuracy with the "handwritten digit" prompt style and around 25.3% accuracy in another CLIP zero-shot MNIST evaluation. Some prompt styles performed slightly better, such as "number only," which achieved around 30.68%, but the overall accuracy remained low.

This shows that CLIP does not automatically generalize well to all visual domains. Even though digits are simple for supervised models trained on MNIST, CLIP was not specifically trained for handwritten digit classification. Therefore, its zero-shot ability depends strongly on whether the target domain resembles the visual-text distribution seen during pretraining.

The MNIST experiment also reinforces the prompt sensitivity issue. Different prompt styles produced different accuracies. This means that when CLIP is used on specialized or out-of-distribution data, prompt engineering can affect performance, but prompt engineering alone may not fully solve the domain mismatch.

Section 2: VideoChat

In this section we compared CLIP and our first Frontier Paper on VideoChat. VideoChat is an end-to-end chat-centric video understanding system that integrates video foundation models with large language models (LLMs). Instead of treating video analysis as a set of isolated vision tasks, it bridges the gap between raw visual perception and complex semantic reasoning. By learning from both video-text pairs and instruction-tuning datasets, the authors created a system capable of spatiotemporal reasoning, event localization, and causal inference, allowing users to converse naturally about the contents of a video.

How VideoChat Improves on CLIP:

Unlike standard vision-language models like CLIP, which process images globally and act like a "bag of words" without an inherent understanding of time or sequence, VideoChat is built for spatiotemporal context. By integrating visual encoders with a generative language decoder, it gains precise spatial perception, state-awareness, and the ability to track causal relationships

(e.g., directionality and intermediate event states). However, as highlighted in the paper and the experiments, scaling this architecture to handle extremely long-term contexts in sequential video processing remains a challenge, as the LLM's attention can become overwhelmed by the sheer number of temporal tokens.

The following experiments 5 -7 Compares CLIP with the VideoChat Architecture.

Experiments 5,6,7: CLIP vs VideoChat

Spatiotemporal Reasoning and Causal Inference

The project also tested CLIP on a synthetic before-and-after image sequence. The image showed a blue circle changing position from the left panel to the right panel. The prompts tested whether CLIP could distinguish between “a blue circle moves from left to right,” “a blue circle moves from right to left,” and “two blue circles are sitting still.”

This task requires temporal reasoning. The model must understand that the left panel represents the earlier state and the right panel represents the later state. CLIP does not naturally process time or sequence; it processes the combined image globally. Therefore, it may recognize that there are two blue circles but fail to infer motion direction correctly.

This experiment highlights a major limitation of image-only vision-language models. They can match objects and scenes to text, but they do not inherently understand temporal order, causal change, or event progression.

Video-Language Model Experiment and Long-Context Failure

The project then moved from static image reasoning to video reasoning. A Video-LLM was tested on a video sequence where a blue square appeared briefly in the first few frames and was then replaced by a red square for the rest of the video.

The model answered that the video showed a red square and that there was no blue square. This was incorrect because the blue square did appear, but only briefly at the beginning. The failure shows that even video-language models can struggle with long temporal context. When most frames show the red square, the model may overweight the dominant later frames and ignore or forget the short initial event.

This result is important because it shows that VideoChat-style models improve over CLIP by adding temporal information, but they are still not perfect. Long videos create many visual tokens, and the model must decide which events are important. Short-lived events can be lost when the model summarizes the overall video.

Therefore, video-language models improve spatiotemporal reasoning, but long-context temporal memory remains a major open challenge.

Section 3: Frontier Paper 2: SigLIP

SigLIP as an Improvement Over CLIP

The project next explored SigLIP, a successor to the CLIP family. SigLIP improves on CLIP by replacing the softmax contrastive loss with a pairwise sigmoid loss. In CLIP, each image is compared against all text captions in a batch using a global softmax. In SigLIP, each image-text pair is treated more independently as a binary matching problem.

This training objective can improve scalability and performance, especially across different batch sizes. SigLIP also tends to perform better on several image-text matching and zero-shot tasks.

The following experiments compare SigLIP, PaliGemma with CLIP architecture performance.

Experiment 9: SigLIP in Object Relationship and Prompting:

We repeated the object relationship experiment with the same data set using SigLIP architecture. It performed much better than “vanilla” CLIP given the prompts. However, the experiments show that SigLIP does not completely solve CLIP’s reasoning limitations. This is because SigLIP still relies on the inherent “ranking the probabilities for output” of CLIP. Thus, SigLIP improves the training objective and can improve performance, but it still inherits some CLIP-like weaknesses in compositional reasoning.

Experiment 10: SigLIP limitation: Multiple-Choice VQA with SigLIP

The project also adapted SigLIP for multiple-choice visual question answering by converting each candidate answer into a text prompt and ranking the answers by image-text similarity.

This approach worked to some extent, but it is not true generative VQA. SigLIP cannot naturally produce an open-ended answer. It can only compare the image with a fixed set of candidate prompts. Therefore, the quality of the answer depends heavily on whether the correct answer is included in the candidate list and how the candidate choices are phrased.

In the dog-cat VQA example, SigLIP predicted “a dog and a cat in the grass” rather than the more relational action choices. This indicates that SigLIP is better at identifying scene-level content than reasoning about precise actions.

This experiment shows that image-text ranking models can be used for multiple-choice VQA, but they are limited compared to models designed for question answering.

Experiment 11: PaliGemma and Generative VQA vs SigLIP

The project discusses PaliGemma as a more appropriate model for true visual question answering. PaliGemma combines a SigLIP vision encoder with a Gemma language decoder. The vision encoder converts the image into visual embeddings, and the language decoder uses those visual tokens plus the text prompt to generate a natural-language answer.

This architecture is different from CLIP and SigLIP. CLIP and SigLIP mainly perform image-text similarity matching, while PaliGemma can generate answers. This makes PaliGemma more suitable for open-ended VQA, image captioning, and instruction-following tasks.

The key interpretation is that better VQA requires more than image-text alignment. It requires a language decoder that can reason over visual features and produce structured textual responses. Therefore, generative vision-language models are a natural next step beyond CLIP-style contrastive models.

SigLIP limitation: Counting Experiment

The counting experiment tested whether models could answer questions such as “How many dogs are in the image?”, “How many cats are in the image?”, and “How many animals are in the image?”. SigLIP produced uncertain results in the counting task. In one ranking setup, it predicted “one dog” with probability around 0.50, while “two dogs” was also close with probability around 0.46. In the multiple-choice VQA setup, it predicted “two dogs” with probability around 0.40. The relatively close probabilities show uncertainty.

A simple ViLT performed better on the explicit counting questions. It answered “2” for dogs, “0” for cats, and “2” for animals, with high confidence values. This suggests that models fine-tuned specifically for VQA can perform better on structured question-answering tasks than pure image-text similarity models.

The interpretation is that counting requires object-level grounding and sometimes explicit supervision. CLIP and SigLIP are not designed primarily for counting, so their predictions can be unstable. VQA-trained models are more appropriate when the task requires direct answers to visual questions.

Experiment 12: SigLIP and CLIP on MNIST

The final MNIST comparison showed that SigLIP performed much better than CLIP on zero-shot handwritten digit classification. SigLIP achieved around 78.6% zero-shot accuracy on 1000 MNIST samples, while CLIP achieved around 25.3% in the comparable zero-shot evaluation.

This is a strong result because it shows that newer vision-language models can improve significantly over CLIP on certain out-of-distribution tasks. However, SigLIP still does not reach the performance of a supervised MNIST classifier, which would typically perform much better when trained directly on digit labels. Therefore, SigLIP improves zero-shot transfer, but task-specific supervised learning is still preferable when high accuracy is required on a specialized dataset.

Overall Interpretation

The experiments support one main conclusion: CLIP is a strong image-text alignment model, but it is not a complete visual reasoning model. It can recognize common objects and match images to broad text descriptions, but it struggles with fine-grained relationships, prompt sensitivity, localization, temporal reasoning, counting, and out-of-distribution data.

The dog-cat experiments show that CLIP and SigLIP can detect objects but may fail to understand relationships between objects. The prompt sensitivity experiments show that small changes in wording can change predictions. The Grad-CAM experiment shows that correct predictions may not always correspond to precise visual grounding. The MNIST experiment shows that zero-shot transfer is limited when the target domain differs from the pretraining distribution.

The Video-LLM experiment shows that adding video understanding helps with temporal reasoning, but long-context video understanding remains difficult. A model may miss short events if they occupy only a small portion of the video. This means that video-language models still need better temporal memory and event localization.

The SigLIP experiments show that improved training objectives can improve performance, especially on tasks like MNIST, but they do not completely solve compositional reasoning. PaliGemma and ViLT show that VQA-specific or generative architectures are more appropriate for question answering because they go beyond simple image-text similarity.

Conclusion

This project demonstrates that different vision-language models have different strengths. CLIP is useful for zero-shot image classification and broad semantic matching. SigLIP improves some zero-shot performance and scalability. PaliGemma-style models are better suited for open-ended VQA because they combine visual encoding with language generation. ViLT performs better on explicit VQA tasks such as counting because it is trained for question answering.

However, none of these models fully solves visual reasoning. Fine-grained relationships, temporal causality, long-context video understanding, localization, and counting remain challenging. The experiments show that future vision-language models need stronger object-level grounding, better compositional understanding, improved temporal memory, and more robust prompt handling.

The overall lesson is that vision-language alignment is not the same as visual understanding. A model may know that an image contains a dog and a cat, but still fail to understand who is chasing whom, where each object is located, how many objects are present, or what happened earlier in a video. These limitations define important directions for future research in multimodal AI.

